

Proof of Dot Product Equivalent Form

Show $\vec{v} \cdot \vec{u} = |\vec{v}||\vec{u}|\cos(\theta)$

Theorem

For any two vectors $\vec{v}$ and $\vec{u}$ and the angle between them θ

$$\vec{v} \cdot \vec{u} = |\vec{v}||\vec{u}|\cos(\theta)$$

Proof

Lemma 1 - Law of Cosines:

Let A, B, and C be the vertices of a triangle and α , β , and γ be the angles at those respective points

(i) $A = B^2 + C^2 - 2BC\cos(\alpha)$

(ii) $B = A^2 + C^2 - 2AC\cos(\beta)$

(iii) $C = A^2 + B^2 - 2AB\cos(\gamma)$

Assume this statement holds true for any points A, B, and C

Lemma 2 - Distance and Magnitude Formula:

Let A and B be two points such that $A, B \in \mathbb{R}^n$ and let vector $\vec{v} = \vec{AB}$

(i) The distance between A and B is $\sqrt{(B_1 - A_1)^2 + (B_2 - A_2)^2 + \dots + (B_n - A_n)^2}$

(ii) The vector between A and B is $\vec{v} = \begin{bmatrix} B_1 - A_1 & B_2 - A_2 & \dots & B_n - A_n \end{bmatrix}$

(iii) The magnitude of the vector $|\vec{v}| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$

Therefore the magnitude of a vector $\vec{v}$ between points A and B is the distance between A and B

Lemma 3 - Magnitude Squared Dot Product Equivalence::

Let $\vec{v}$ be a vector such that $\vec{v} \in V^n$

(i) $|\vec{v}| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$ by Lemma 2

(ii) $\vec{v} \cdot \vec{v} = v_1 \cdot v_1 + v_2 \cdot v_2 + \dots + v_n \cdot v_n = v_1^2 + v_2^2 + \dots + v_n^2$

(iii) $|\vec{v}|^2 = v_1^2 + v_2^2 + v_n^2$

Therefore $|\vec{v}|^2 = \vec{v} \cdot \vec{v}$

Let A, B, C be three points such that

$\bullet \ A, B, C \in \mathbb{R}^n$

$\bullet \ \vec{v} = \vec{AB} = \begin{bmatrix} B_1 - A_1 & B_2 - A_2 & \dots & B_n - A_n \end{bmatrix}$

$\bullet \ \vec{u} = \vec{AC} = \begin{bmatrix} C_1 - A_1 & C_2 - A_2 & \dots & C_n - A_n \end{bmatrix}$

$\bullet \ \vec{w} = \vec{BC} = \begin{bmatrix} C_1 - B_1 & C_2 - B_2 & \dots & C_n - B_n \end{bmatrix}$

(i) $\vec{u} - \vec{v} = \begin{bmatrix} C_1 - A_1 + A_1 - B_1 & C_2 - A_2 + A_2 - B_2 & \dots & C_n - A_n + A_n - B_n \end{bmatrix} = \begin{bmatrix} C_1 - B_1 & C_2 - B_2 & \dots & C_n - B_n \end{bmatrix} = \vec{w}$

Therefore $\vec{u} - \vec{v} = \vec{w}$

(ii) $|\vec{u} - \vec{v}|^2 = (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v})$ by Lemma 3

(iii) $(\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) = \vec{u} \cdot \vec{u} + \vec{v} \cdot \vec{v} - 2(\vec{v} \cdot \vec{u})$ by distributive property of vectors

(iv) $|\vec{u} - \vec{v}|^2 = |\vec{v}|^2 + |\vec{u}|^2 - 2|\vec{v}||\vec{u}|\cos(\theta)$ by Lemma 1

(v) $|\vec{v}|^2 + |\vec{u}|^2 - 2|\vec{v}||\vec{u}|\cos(\theta) = |\vec{u}|^2 + |\vec{v}|^2 - 2(\vec{v} \cdot \vec{u}) \iff |\vec{v}||\vec{u}|\cos(\theta) = \vec{v} \cdot \vec{u}$

Therefore for any two vectors $\vec{v}$ and $\vec{u}$ and the angle between them θ

$$\vec{v} \cdot \vec{u} = |\vec{v}||\vec{u}|\cos(\theta)$$

is proven true by direct proof